Basics of Counting

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Counting

- Counting problems arise throughout maths and computer science
- Suppose you have come up with the following rule for creating passwords:
  - it should have $6, 7$ or 8 characters
  - A character is either a digit$(0..9)$, a letter$(a..z$ and $A..Z)$ or an underscore.
  - A password must contain at least one digit.
- How many such passwords are there?
- Techniques needed to answer this kind of question and a variety of other counting problems will be introduced.

Basic Counting Principles

Basic counting principles serve as the foundation for almost all counting techniques. Two basic counting principles.

- The product rule
- The sum rule

The product rule - definition

Definition

Suppose a procedure can be broken down into a sequence of two(2) tasks. If there are n_1 ways of doing the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 \times n_2$ ways of doing the procedure

Example

Example

A new company has two employees John and Mary. The company rents an office building with 10 offices

How many ways are there to assign different offices to these two employees?

Solution

$\Rightarrow$ If we start with John, there are 10 different ways to assign offices to John.

$\Rightarrow$ Once John has been assigned an office, there are 9 different ways to assign an office to Mary that has not been assigned to John.

$\Rightarrow$ Using the product rule, there are $10 \times 9 = 90$ different ways to assign offices to these two employees.

Exercise

Example

The chairs of a Lecture hall are to be labeled with a letter followed by a positive integer not exceeding 100.

What is the largest number of chairs that can be labeled differently?

Solution

$\Rightarrow$ 26 different ways to select a letter

$\Rightarrow$ 100 ways to select a positive integer not exceeding 100.

$\Rightarrow$ Using the product rule, $26 \times 100 = 2600$ chairs can be labeled.

Example

Example

How many different bit strings of length 7 are there?

Solution

$\Rightarrow$ The 1st bit string can be choosen in two different ways (0 or 1)

$\Rightarrow$ each of the 2nd,3rd,4th,5th,6th, and 7th bit string can be selected in two different ways.

$\Rightarrow$ Therefore each of the seven bit string can be selected in 2 different ways.

$\Rightarrow$ By the product rule, there are: $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$ bit strings of length 7

Example

Example

How many number plates are available if each number plate begins with a fixed letter (B) followed by 3 digits and finally ending with 3 letters. No sequence of numbers and letters are prohibited. So $B000AAA$ is a legal number plate.

Solution

$\Rightarrow$ Since B is fixed, 1 way to select B.

$\Rightarrow$ for each of the 3 digits there are 10 ways to select each, giving $10 \times 10 \times 10$ different ways

$\Rightarrow$ For each of the 3 letters, there are 26 ways to select each, giving $26 \times 26 \times 26$ different ways.

$\Rightarrow$ So total is: $1 \times 10 \times 10 \times 10 \times 26 \times 26 \times 26 = 17,576,000$ number plates.

Example

Example

For simplicity, assume Mascom is the only Mobile service provider with numbers that begin with 71. Note that this might not be true in real life. Each number has 8 digits including the fixed starting digits.
How many Mascom numbers beginning with 71 **are there?**

Solution

$\Rightarrow$ Each of the 6 digits after 71 can be choosen in 10 different ways. Note that this will include the number 71000000

$\Rightarrow$ Using the product rule, there are:
$1 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 1,000,000$ numbers that start with a 71.

The sum rule

Definition

If a task can be done in n_1 ways and a second task in n_2 ways, and if these tasks cannot be done at the same time, there are $n_1 + n_2$ ways to do the task.

Example

Example

Suppose that either a Computer Science or Physics student is to be chosen to represent science students in some UB committee. Assume there are 150 CS students and 240 Physics students.
How many different choices are there for this representative. No Student is both a Physics and Computer Science student.

Solution

Since choosing a Physics student if different from choosing a CS student. By the SUM rule there are $240 + 150 = 390$ different ways to select the representative.

Example

Example

A student can choose a CS project from one of 3 lists: L_1, L_2, L_3. L_1 has 23 projects, L_2 has 15 projects and L_3 has 19 projects. No project is in more than one list. How many ways are there to choose a project?

Solution

A student can select a project from L_1 or L_2 or L_3. Since no project is in more than 1 list, there are $23 + 15 + 19 = 57$ ways to choose a project.

Sum rule - extended version

Note that the previous example has 3 tasks instead of 2. We can generalize the sum rule to:

Definition

Suppose a task can be done in one of n_1 ways or in n_2 ways, ... , or one of n_m ways where none of the set of n_i is the same as any of the n_j ways for $1 \leq i \leq j \leq m$, THEN the number of ways to do the task is:
$n_1 + n_2 + ... + n_m$

More Complex problems

Problems that require to be solved using a combination of the SUM and PRODUCT rule.

Example

The syntax of variable names in an old programming language called BASIC is as follows:

- A variable has one(1) or two alphanumeric characters where uppercase and lowercase letters are not distinguished.
- Alphanumeric means: the 26 letters and the 10 digits
- A variable name MUST begin with a letter and must be different from 5 strings that are reserved. In BASIC, examples of reserved words are: for, if, etc. These reserved words cannot be used as variable names.

How many different variable names are there on this version of BASIC?

Solution

- Let V_1 be the number of 1 character variables. Since a variable MUST start with a letter, one character variables are made up of the 26 letters. so $V_1 = 26$
- Let V_2 be the number of 2 character variables. These are formed starting with any letter, followed by any letter of digit. So $V_2 = 26 * 36$.
- Total number of variables using the sum rule is: $V_1 + V_2 - 5 = 26 + (26 * 36) - 5 = 957$
- -5 to exclude the 5 reserved words

Example

Example

Structure of passwords:

- Each password has 6 to 8 characters
- A character is an uppercase letter or a digit
- Each password has at least one digit

How many possible passwords are there?

Solution

- Let T be the total number of passwords.
- Let T_6 be the number of 6 character passwords.
- Let T_7 be the number of 7 character passwords.
- Let T_8 be the number of 8 character passwords.
- By the sum rule: $T = T_6 + T_7 + T_8$

Solution contd. - find T_6

- Find the total number of 6 character strings with no digit: Using the product rule its: $26 \times 26 \times 26 \times 26 \times 26 \times 26 = 26^6$
- Find the total number of 6 character strings with digits: $(26 + 10)^6 = 36^6$
- Total number of 6 character strings excluding those with no digits: $36^6 - 26^6 = 1,867,866,560$
- So $T_6 = 1,867,866,560$

Solution contd. - find T_7

- Find the total number of 7 character strings with no digit: Using the product rule its: $26 \times 26 \times 26 \times 26 \times 26 \times 26 \times 26 = 26^7$
- Find the total number of 7 character strings with digits: $(26 + 10)^7 = 36^7$
- Total number of 7 character strings excluding those with no digits: $36^7 - 26^7 = 70,332,353,920$
- So $T_7 = 70,332,353,920$

Solution contd. - find T_8

- Find the total number of 8 character strings with no digit: Using the product rule its: $26 \times 26 \times 26 \times 26 \times 26 \times 26 \times 26 \times 26 = 26^8$
- Find the total number of 8 character strings with digits: $(26 + 10)^8 = 36^8$
- Total number of 8 character strings excluding those with no digits: $36^8 - 26^8 = 2,612,282,842,880$
- So $T = T_6 + T_7 + T_8 = 2,684,483,063,360$

The inclusion-exclusion principle

- suppose that a task can be done in n_1 ways or in n_2 ways
- Some of the set of n_1 ways are similar to the set of n_2 ways
- Calculating the number of ways to do the task by adding $(n_1 + n_2)$ leads to an over count.
- Let the similar ways be n_s ways.
- Then the total number of ways to do the task is $n_1 + n_2 - n_s$

Example

Example

How many bit strings of length 8 either start with a 1 or end with two bits 00?

Solution

- $A_1 =>$ Number of bit strings of length 8 starting with 1: $= 2^7$
- $A_2 =>$ Number of bit strings of length 8 ending with 00: $= 2^6$
- $A_s =>$ Common bit strings. Those that begin with 1 and end with 00: $= 2^5$
- So the total number of bit strings that start with a 1 or end with 00: $= A_1 + A_2 - A_s$
- Total: $2^7 + 2^6 - 2^5$

Inclusion-exclusion principle - in terms of sets

- let A_1 be a set. Number of ways to select elements from it is $|A_1|$
- let A_2 be a set. Number of ways to select elements from it is $|A_2|$
- Number of ways to select elements from A_1 or A_2 is the number of ways to select elements from their union. $|A_1 \cup A_2|$
- where, $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$

Using tree diagrams in counting

- Structure of a tree:
  - It has a root node
  - A number of branches leaving the root node
  - Additional branches leaving the endpoints of other branches
  - Leaves are endpoints of branches not having other branches leaving them
- How trees are used in counting:
  - A branch represents each possible choice
  - Leaves represent possible outcomes

Example

Example

How many bit strings of length four do not have consecutive ones? List them.

Exercise

Example

Example

Suppose that you want to print t-shirts for CS and sell to your classmates. The t-shirts come in five different sizes: S, M, L, XL and XXL. Further suppose that each size comes in four colors: white, red, green and black, except for XL, which comes only in red, green, and black, and XXL which comes only in green and black. How many different shirts do you have to stock to have at least one of each available size and color of the t-shirt?

Exercise

Questions

Questions?